MANAS REDDY ARUMALLA

Robotics & Control Engineer | MSc Autonomy Technologies, FAU Erlangen-Nürnberg

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SUMMARY

Robotics engineer specializing in **control systems, motion planning, and sim-to-real deployment**. Published author (ICRM 2025) on hopping-robot dynamics and contributor to a DST-funded research project. Hands-on across the full stack — dynamic modelling and controller design in MATLAB, validation in MuJoCo/Gazebo, and deployment to ROS 2 and embedded hardware. Research-grade portfolio including a vision-only drone that lands on moving ship decks and a bimanual manipulation platform.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg *2026 – Present*

M.Sc. Autonomy Technologies — Nürnberg, Germany

Amrita Vishwa Vidyapeetham *2021 – 2025*

B.Tech, Automation & Robotics — Coimbatore, India | CGPA 8.84/10.0 (First Class with Distinction)

Relevant coursework: Control Systems, Robotics & Mobile Robotics, Computer Vision, Machine Learning, Industrial Automation, Embedded Systems.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Control & Estimation: LQR, MPC, SMC, PID, backstepping, H_∞, adaptive & predictive control; state-space modelling; trajectory tracking (Pure Pursuit, Stanley, DWA)

Planning & Navigation: A*, RRT/RRT*, PRM, Hybrid A*, APF; RVO multi-robot avoidance; PDDL / PlanSys2; ROS 2 Nav2

ML / RL: Reinforcement Learning (PPO), genetic-algorithm optimization, computer vision (OpenCV, MediaPipe)

Robotics Frameworks: ROS / ROS 2, Gazebo, MuJoCo, CoppeliaSim, PX4 / MAVLink

Simulation & CAD: Fusion 360 / Inventor, Adams, Simscape, Simulink

Embedded & Hardware: Arduino, Teensy, Raspberry Pi, microcontroller programming, Embedded C, sensor integration

Operating Systems: Linux, Windows

PROFESSIONAL EXPERIENCE

Robotics Engineer, Humanoid Robotics Startup (Stealth) *Feb 2026 – Apr 2026*

- Built robotic-manipulation simulations in Gazebo and implemented PDDL task planning, letting task sequences be validated in simulation before committing to hardware.
- Simulated and tested a K-Scale humanoid in ROS 2 and stood up the sim-to-hardware deployment pipeline.

Research Assistant, Amrita Vishwa Vidyapeetham (under Dr. Rammohan Sriramdas) *Sep 2025 – Jan 2026*

- Developed a DST-funded two-wheeled hopping robot — characterized its jump dynamics and designed the stabilizing control, work published at ICRM 2025.
- Enhanced the control architecture and navigation performance of a quadruped mobile robot.
- Authored three research papers — one published, two under review.

Robotics Intern, Anvi Robotics *Jun 2024 – Aug 2024*

- Designed a water-cleaning unmanned surface vehicle (USV): component selection, CAD, and ROS simulations integrating MAVLink with PX4.
- Integrated vision-based trash detection and autonomous navigation in ROS, enabling the USV to locate and route to floating trash on its own.

Robotics Intern, Sony SSUP Program *Oct 2023 – Dec 2024*

- Co-developed a four-legged mobile robot for farm assistance and cattle health monitoring under the Sony SSUP program.
- Programmed and integrated multiple sensors and control algorithms, and troubleshot hardware to keep experiments running with minimal downtime.

Robotics Intern, 7s Technologies *Mar 2024 – Apr 2024*

- Designed an AR/VR motion-simulation chair with real-time control; selected components and ran load/performance calculations to hit motion-fidelity targets.

SELECTED PROJECTS

Vision-Based Drone Landing on a Moving Platform *Python, MuJoCo, CasADi*

- Built a **vision-only autonomous quadrotor** that lands on moving ground rovers and 6-DOF seakeeping ship decks from onboard sensors alone. ArUco fiducials fused in a relative-state EKF with a stabilized gimbal, no ground-truth state in the loop.
- Designed the control and safety stack: **geometric SE(3)** and nonlinear / tube-DOB MPC (CasADi), a **Hamilton–Jacobi reachability safety shield**, and higher-order control barrier functions for sense-and-avoid.
- Added reinforcement-learning (PPO / LSTM-PPO) landing policies and extended the system to a **decentralized multi-drone swarm** with Kalman-consensus estimation and a CBF-QP safety filter.

OpenArm — Bimanual Manipulation Platform *Python, MuJoCo*

- Built a control, planning, and learning stack for the **bimanual Enactic OpenArm** in MuJoCo: forward/inverse kinematics, Cartesian and **compliant (admittance) control**, and RRT-Connect obstacle avoidance.
- Implemented a range of manipulation skills — pick-and-place, articulated-object handling (drawers/valves), cloth folding, collision-aware bimanual hand-over, and dynamic ball-catching via **Kalman + MPC interception** — plus a learned ACT vision policy and an RL insertion suite.

Comparative Control Study — Two-Wheeled Inverted Pendulum *MATLAB, MuJoCo, python-control*

- Derived and benchmarked **15+ controllers** (LQR, MPC, H_∞, SMC, backstepping, NDI, LPV, adaptive, predictive) for a self-balancing Segway-type robot, with every controller **auto-tuned by optimization algorithms**.
- Added energy-based swing-up and realistic state estimation, and validated each design in a MuJoCo simulation against a MATLAB state-space model.

navstack — Mobile-Robot Navigation & Control Platform *Python, MuJoCo*

- Built a modular navigation stack: planners (A*, Dijkstra, RRT/RRT*, PRM, PSO, APF, semantic & kinodynamic Hybrid A*), a MuJoCo simulator across **6 drive types** (diff 2WD/4WD, mecanum, omni, Ackermann, bicycle), and trajectory controllers (Pure Pursuit, Stanley, DWA, MPC).
- Implemented **RVO/Velocity-Obstacle multi-robot swarm avoidance** and a PPO reinforcement-learning controller for a self-balancing Segway.

TurtleBot3 Autonomous Task Planning *ROS 2 Humble, PlanSys2, PDDL, Nav2*

- Built a goal-driven planner that lets a TurtleBot3 derive its own optimal navigation-and-cleaning action sequence from a PDDL world description, with Nav2 integration and failure-recovery testing.

Pharma Bot — Autonomous Medicine Delivery *OpenCV, CoppeliaSim, Raspberry Pi*

- Developed a vision-guided delivery robot — OpenCV arena perception, graph-based shortest-path planning, a CoppeliaSim digital twin, then real Raspberry Pi hardware. **Qualified to Level 3** in the national e-Yantra Robotics Competition (IIT Bombay).

Additional: Steady Stride — physical self-balancing robot with onboard manipulator (Arduino/C++, PID & LQR, IMU sensor fusion); manipdyn 6-DOF UR5e control & planning lab (8 controllers, 5 planners); quadrotor control benchmarks in Python & MATLAB (PID / LQR / SE(3) / SMC / MPC, GA-tuned); real-time hand-tracking teleoperation of a Shadow Hand & Crazyflie (MediaPipe).

PUBLICATIONS & RESEARCH

- M. R. Arumalla, R. Sriramdas, V. R. Devarakonda, A. Vincent, *"Mapping the Dynamics of Robotic Jumps in Hopping Robots,"* International Conference on Robotics and Mechatronics (ICRM 2025), 2025.
- Two additional manuscripts currently under review (2026).

ACHIEVEMENTS & LEADERSHIP

- **e-Yantra Robotics Competition 2022-23** (IIT Bombay) — awarded Level 3 for an autonomous pharma-delivery bot.
- **Robotics Club** — led the projects team; designed and delivered a hands-on ROS workshop.
- **NSS Volunteer** — contributed to community camps and social-outreach initiatives.